Statistics 651
Hierarchical Modeling

Winter 2026

BYU
Stats

Objectives

1. Formulate hierarchical models to appropriately model stratified data
2. Fit hierarchical models via MCMC
3. Draw from predictive distributions at appropriate population levels
4. Assess whether exhangeability is reasonable in a given scenario
5. Express models for dependent data hierarchically or marginally

Resources

- BDA3: Ch 5
- BIDA: Ch 4.12–4.13
- Hoff: Ch 8.2–8.5 (great treatment of the normal hierarchical model)

Stratified data

Motivation

Example

In a recent Utah Jazz game[1], ten players attempted free throws.

Results: (# made / # attempts)

15/20, 4/7, 3/6, 4/4, 2/4, 10/12, 5/6, 10/12, 2/2, 0/1

Suppose we are interested in:

- Estimating a typical NBA player's free-throw ability
- Predicting a typical NBA player's free-throw performance
- Estimating the free-throw ability of the fourth player listed above
- Predicting the free-throw performance of the fourth player listed above

Motivation

Example (continued)

Free throws: (# made / # attempts)

15/20, 4/7, 3/6, 4/4, 2/4, 10/12, 5/6, 10/12, 2/2, 0/1

If estimating/predicting for a typical NBA player:

Possible strategies

- $\dfrac{15+4+3+\dots+2+0}{20+7+6+\dots+2+1}$ completely pooled estimate

- $\left(\frac{15}{20}\right)\times\frac{1}{10}+\left(\frac{4}{7}\right)\times\frac{1}{10}+\dots+\left(\frac{0}{1}\right)\times\frac{1}{10}$ equally weighted combination of individual estimates

Motivation

Example (continued)

Free throws: (# made / # attempts)

15/20, 4/7, 3/6, 4/4, 2/4, 10/12, 5/6, 10/12, 2/2, 0/1

If estimating/predicting for the fourth listed player:

What is the maximum-likelihood estimate under a Binomial model? How would you predict??

$$\hat{\theta}_4^{MLE} = \frac{4}{4} = 1$$ "perfect free throw shooter"

Would it be appropriate to incorporate the other players' data to influence your estimate/predictions?

How would you do this?

Create an informed Beta prior for θ_4 using other nine players' data?

Motivation

Example (continued)

Free throws: (# made / # attempts)

15/20, 4/7, 3/6, 4/4, 2/4, 10/12, 5/6, 10/12, 2/2, 0/1

Estimate/predict for:

- A typical NBA player.
- The fourth player listed above.

Could we address **both** objectives with a single model?

Motivation

Example (continued)

Free throws: (# made / # attempts)

15/20, 4/7, 3/6, 4/4, 2/4, 10/12, 5/6, 10/12, 2/2, 0/1

Estimate/predict for:

- A typical NBA player.
- The fourth player listed above.

Model: $y_i \mid \theta_i \overset{\text{ind}}{\sim} \text{Binom}(n_i,\, \theta_i)$

What should the prior for θ_i look like?

$\theta_i \mid \alpha, \beta \overset{\text{iid}}{\sim} \text{Beta}(\alpha, \beta)$

$\alpha = M\mu \qquad \beta = M(1-\mu)$

$\mu \sim \text{Beta}(a_\mu, b_\mu) \qquad M \sim \text{Gamma}(a_M, b_M)$

Hierarchical models

Hierarchical models

Model formulation

Generic model form

$$\begin{aligned} \boldsymbol{y} \mid \boldsymbol{\theta} &\sim f(\boldsymbol{y}|\boldsymbol{\theta}) \,, \\ \boldsymbol{\theta} \mid \boldsymbol{\phi} &\sim \pi(\boldsymbol{\theta}|\boldsymbol{\phi}) \,, \\ \boldsymbol{\phi} &\sim \pi_0(\boldsymbol{\phi}) \,, \end{aligned}$$

where f is the **sampling model** with **parameters** $\boldsymbol{\theta}$ and **hyperparameters** $\boldsymbol{\phi}$.

A hierarchical model can be composed of as many levels as necessary.

It is common for $\boldsymbol{y}$ to be conditionally independent of $\boldsymbol{\phi}$ (conditional on $\boldsymbol{\theta}$).

Joint distribution:

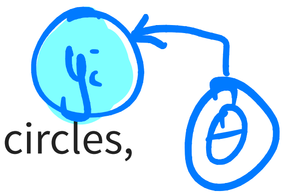
$$p(y, \theta, \phi) = \pi_0(\phi)\pi(\theta|\phi) f(y|\theta)$$

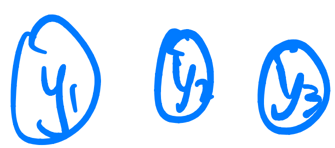
Hierarchical models

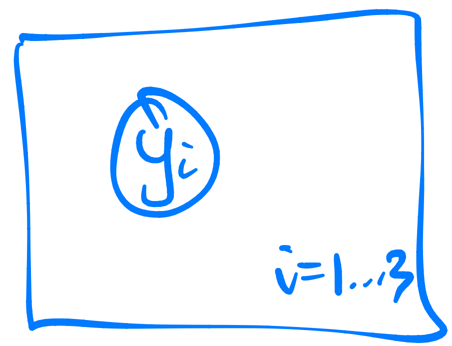
Graphical representation

Hierarchical models can be visually represented using **directed acyclic graphs** (DAGs), where

- observed stochastic nodes (data) are represented with filled circles,
- unobserved stochastic nodes (parameters) are represented with open circles,
- dependence relationships are represented with arrows connecting nodes, and
- replication is represented with plates.







Hierarchical models

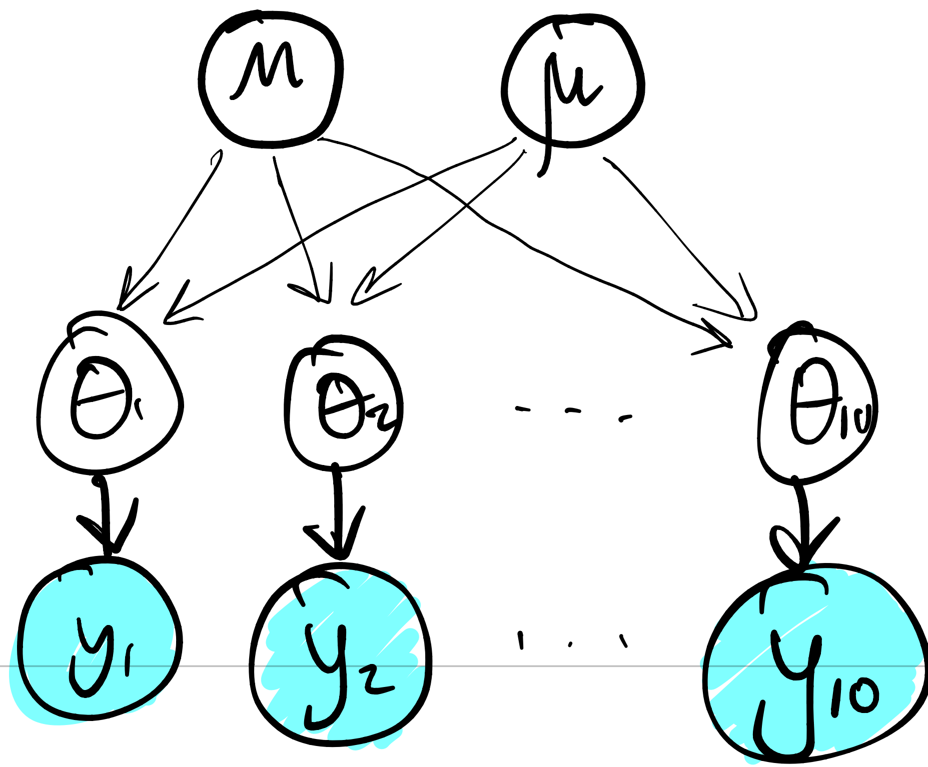
DAG example

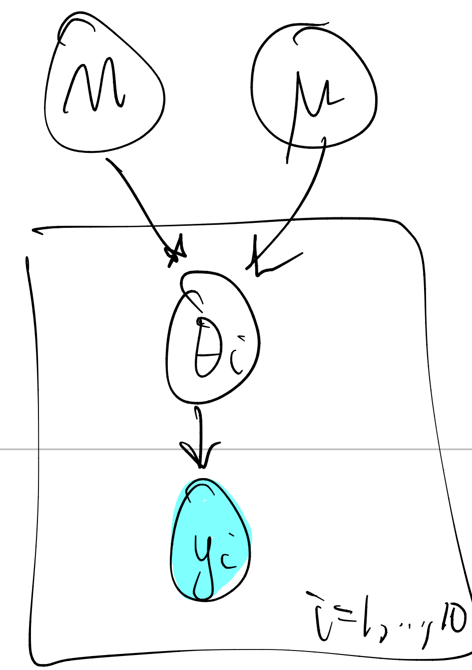
Draw a DAG for the hierarchical Binomial free-throw model.

$$y_i \mid \theta_i \overset{ind}{\sim} \text{Binom}(n_i, \theta_i)$$

$$\theta_i \mid M, \mu \overset{iid}{\sim} \text{Beta}(M\mu, M(1-\mu))$$

$$M \sim Ga, \quad \mu \sim Beta$$





Hierarchical models

Marginal model

The hierarchical model

$$\boldsymbol{y} \mid \boldsymbol{\theta} \sim f(\boldsymbol{y}|\boldsymbol{\theta})\,,$$
$$\boldsymbol{\theta} \mid \boldsymbol{\phi} \sim \pi(\boldsymbol{\theta}|\boldsymbol{\phi})$$
$$\boldsymbol{\phi} \sim \pi_0(\boldsymbol{\phi})$$

is equivalent to modeling

$$f_{\text{marg}}(\boldsymbol{y}|\boldsymbol{\phi}) = \int f(\boldsymbol{y}|\boldsymbol{\theta})\,\pi(\boldsymbol{\theta}|\boldsymbol{\phi})\,\mathrm{d}\boldsymbol{\theta}$$

directly.

Joint distribution of $(\boldsymbol{y}, \boldsymbol{\phi})$:

$$p(y, \phi) = f_{marg}(y|\phi)\,\pi(\phi)$$

Hierarchical models

Joint posterior distribution

can also be written as

$$\pi(\boldsymbol{\theta}, \boldsymbol{\phi} \mid \boldsymbol{y}) = \frac{p(y, \theta, \phi)}{\iint p(y, \theta, \phi)\, d\theta\, d\phi} = \pi(\theta \mid \phi, y)\, \pi(\phi \mid y)$$

MCMC can proceed by targeting $\pi(\boldsymbol{\theta}, \boldsymbol{\phi} \mid \boldsymbol{y})$ all at once (as STAN does using HMC) or by Gibbs sampling.

Sometimes it is a good idea to sample the **collapsed** posterior, if possible:

can also be written as

$$\pi(\boldsymbol{\phi} \mid \boldsymbol{y}) = \int \pi(\theta, \phi \mid y)\, d\theta = \frac{\pi(\theta, \phi \mid y)}{\pi(\theta \mid \phi, y)}$$

Hierarchical models

Conditional independence and DAGs

DAGs are also useful because conditional-independence structures become apparent by "moralizing" the graph:

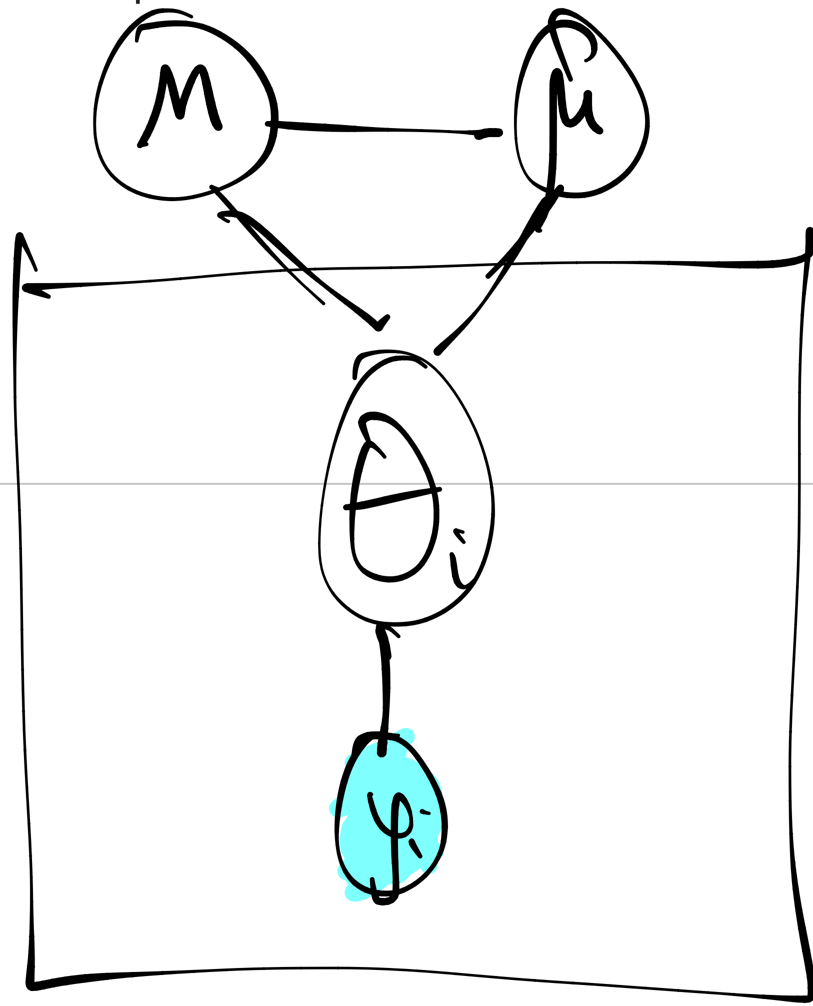
1. Two nodes sharing a common immediate descendant are connected with undirected edges.
2. Directional arrows are replaced with undirected edges.
3. Nodes that are separated by common connections with other nodes are independent, conditional on the connecting nodes.

Where would conditional independence be useful?

Hierarchical models

Free-throw example

Show the undirected conditional independence structure for the Binomial free-throw model.



Hierarchical models

Example (continued)

Free throws: (# made / # attempts)

15/20, 4/7, 3/6, 4/4, 2/4, 10/12, 5/6, 10/12, 2/2, 0/1

Fit the model

$$
\begin{aligned}
y_i \mid \theta_i &\overset{\text{ind}}{\sim} \text{Binom}(n_i, \theta_i)\,, \quad i = 1, \ldots, n, \\
\theta_i \mid \alpha, \beta &\overset{\text{iid}}{\sim} \text{Beta}(\alpha, \beta)\,, \quad i = 1, \ldots, n, \\
(\alpha, \beta) &\sim \Pi_{\alpha,\beta}\,,
\end{aligned}
$$

M, μ

using

- a full Gibbs sampler — Sample $\pi(M, \mu, \{\theta_i\} \mid y)$ using full conditionals
- a collapsed Gibbs sampler — sample $\pi(M, \mu \mid y)$
- a PPL

$$p(y, \underset{\sim}{\theta}, M, \mu) = \prod_{i=1}^{n} \left[\binom{n_i}{y_i} \theta_i^{y_i} (1-\theta_i)^{n_i - y_i} \right] \times$$

$$\prod_{i=1}^{n} \left[\frac{\Gamma(M)}{\Gamma(M\mu)\Gamma(M(1-\mu))} \theta_i^{M\mu - 1} (1-\theta_i)^{M(1-\mu)-1} \right] \times$$

$$\frac{\Gamma(a_\mu + b_\mu)}{\Gamma(a_\mu)\Gamma(b_\mu)} \mu^{a_\mu - 1} (1-\mu)^{b_\mu - 1} \frac{b_m^{a_m}}{\Gamma(a_m)} M^{a_m - 1} e^{-\frac{M}{b_m}} \times$$

$$\mathbb{1}\left((\{y_i\}, \{\theta_i\}, M, \mu) \in S \right)$$

Full Gibbs Sampler

$$p(\theta_i \mid \cdots) \propto \theta_i^{y_i}(1-\theta_i)^{n_i-y_i}\,\theta_i^{M\mu-1}\,(1-\theta_i)^{M(1-\mu)+}$$

$$\propto Be\big(M\mu + y_i,\; M(1-\mu)+n_i-y_i\big)$$

$$p(M \mid \cdots) \propto \prod_{i=1}^{n}\left[Be(\theta_i \mid M)\right] Ga(M)$$

$$p(\mu \mid \cdots) \propto \prod_{i=1}^{n}\left[Be(\theta_i \mid \mu)\right] Be(\mu)$$

Collapsed Gibbs Sampler

$$\int p(y, \underset{\sim}{\theta}, M, \mu)\, d\underset{\sim}{\theta} = Ga(M) \times Be(\mu) \prod_{i=1}^{n} \int Bin(y_i|\theta_i)\, Be(\theta_i)\, d\theta_i$$

$$\binom{n_i}{y_i} \frac{\Gamma(M)}{\Gamma(M\mu)\Gamma(M(1-\mu))} \frac{\Gamma(M\mu + y_i)\Gamma(M(1-\mu) + n_i - y_i)}{\Gamma(M + n_i)} \propto \frac{B(M\mu + y_i, M(1-\mu) + n_i - y_i)}{B(M\mu, M(1-\mu))}$$

So $$p(M|\mu, y) \propto \frac{B(M\mu + y_i, M(1-\mu) + n_i - y_i)}{B(M\mu, M(1-\mu))} \times Ga(M)$$

and $$p(\mu|M, y) \propto \underbrace{\qquad\qquad}_{\text{this}} \times Beta(\mu)$$

Predictive distributions

Predictive distributions

Example (continued)

Recall that our original objectives in the free throw example were to:

- Estimate a typical NBA player's free-throw ability
- Predict a typical NBA player's free-throw performance
- Estimate the free-throw ability of the fourth player (4/4)
- Predict the free-throw performance of the fourth player (4/4)

Use the model to address each.

Predictive distributions

Example (continued)

1. Estimate the free-throw ability of the fourth player (4/4).

Which quantity do we use here? Compare prior and posterior for this quantity.

θ_4

Predictive distributions

Example (continued)

2. Predict the performance of the fourth player (4/4) on his next five free throws.

Which quantity do we use here? Compare prior and posterior for this quantity.

$$y_4^{pred} \mid \theta_4 \sim \text{Binom}(5, \theta_4)$$

$$p(y_4^{pred} \mid \text{data}) = \int_0^1 \binom{5}{\theta_4} \theta_4^{y_4^{pred}} (1-\theta_4)^{5-y_4^{pred}} \pi(\theta_4 \mid \text{data})\, d\theta_4$$

See code – just draw y_4^{pred} for each posterior sample of θ_4

Predictive distributions

Example (continued)

3. Estimate a typical NBA player's free-throw ability.

Which quantity do we use here? Compare prior and posterior for this quantity.

$$\theta^{new} \sim \text{Beta}(M\mu,\ M(1-\mu))$$

$$p(\theta^{new} \mid \text{data}) = \int_0^1 \int_0^\infty \text{Beta}(\theta^{new} \mid M, \mu)\, \pi(M, \mu \mid \text{data})\, dM\, d\mu$$

Beta density

See code — just draw θ^{new} from the Beta distribution above for each posterior sample of M and μ.

Predictive distributions

Example (continued)

4. Predict the performance of a typical NBA player on his next five free throws.

Which quantity do we use here? Compare prior and posterior for this quantity.

$$y^{new} \mid \theta^{new} \sim \text{Binom}(5, \theta^{new})$$
$$\theta^{new} \sim \text{Beta}(M\mu, M(1-\mu))$$

$$p(y^{new} \mid \text{data}) = \int_0^1 \binom{5}{y^{new}} (\theta^{new})^{y^{new}} (1-\theta^{new})^{5-y^{new}} \underbrace{\pi(\theta^{new} \mid \text{data})}_{\text{from previous slide}} \, d\theta^{new}$$

See code – first draw θ^{new} as in prev. slide, then draw y^{new} from Binom using θ^{new}

Predictive distributions

What happens if we just plug in $\mathbb{E}(\theta^* \mid \alpha, \beta)$ for θ^* when estimating the predictive distribution of y^*?

$$\mathrm{Var}(y^*) = \mathbb{E}_{\theta^*}\Big[\underbrace{\mathrm{Var}(y^* \mid \theta^*)}_{5\theta^*(1-\theta^*)}\Big] + \cancel{\mathrm{Var}_{\theta}\Big[\mathbb{E}(y^* \mid \theta^*)\Big]}$$

Exchangeability

Exchangeability

Example (continued)

Free throws: (# made / # attempts)

15/20, 4/7, 3/6, 4/4, 2/4, 10/12, 5/6, 10/12, 2/2, 0/1

Is there any reason we should consider a θ_i NOT interchangeable with the others in the model?

– Knowing players ID, covariate info

Exchangeability

Example

Five participants in a survey were asked if they were generally happy.

Let $X_i = 1$ if they responded positively and 0 otherwise.

Suppose we must assign probabilities to each of the following sequences:

$$\begin{aligned} \mathbb{P}(1,1,0,0,1) &= ? \\ \mathbb{P}(1,0,1,0,1) &= ? \\ \mathbb{P}(1,0,1,1,0) &= ? \end{aligned}$$

Is there any argument for assigning distinct values?

Exchangeability

Definition

A sequence of random variables, $X_1, X_2, \ldots$ with $X_i \in \mathcal{X}$ is considered **exchangeable** if its joint distribution is invariant to permutation of the indices. That is,

$$\mathbb{P}(X_1 \in A_1, X_2 \in A_2, \ldots) = \mathbb{P}(X_{\sigma(1)} \in A_1, X_{\sigma(2)} \in A_2, \ldots)$$

for all $A_i \in \mathcal{B}(\mathcal{X})$, where $\sigma(\cdot)$ is any function that permutes the indexes.

Exchangeability

Example (continued)

Five participants in a survey were asked if they were generally happy.

Let $X_i = 1$ if they responded positively and 0 otherwise.

Consider the following:

$$\mathbb{P}(X_5 = 1) = a$$
$$\mathbb{P}(X_5 = 1 \mid X_1 = 1, \ldots, X_4 = 1) = b$$

Should we have $a = b$, $a > b$, or $a < b$?

If $a \neq b$, then X_5 is not independent of $X_1, \ldots, X_4$.

But they can still be exchangeable
(a and b must not change if
we swap indices on the Xs)

Exchangeability

Example (continued)

Five participants in a survey were asked if they were generally happy.

Let $X_i = 1$ if they responded positively and 0 otherwise.

Now suppose we know the “rate of happiness” $\theta \in [0, 1]$ in the population and that

$$\mathbb{P}(X_i = 1 \mid \theta) = \theta\,,$$
$$\mathbb{P}(X_i = 1 \mid X_{j \neq i}, \theta) = \theta\,.$$

Then $\{X_i\}$ are **conditionally independent** given θ and

$$\mathbb{P}(X_1 = x_1, \ldots, X_5 = x_5 \mid \theta) = \theta^{\sum x_i}(1 - \theta)^{5 - \sum x_i}\,.$$

Exchangeability

Example (continued)

Five participants in a survey were asked if they were generally happy.

Let $X_i = 1$ if they responded positively and 0 otherwise.

If Π represents our beliefs about θ, then the marginal distribution of $X_1, \ldots, X_5$ is

$$\mathbb{P}(X_1 = x_1, \ldots, X_5 = x_5) = \int \underbrace{p(x_1, x_2, \ldots, x_5 \mid \theta)}_{\prod \theta^{x_i}(1-\theta)^{1-x_i}} \Pi(d\theta)$$

if we assume conditional independence

and

$$\mathbb{P}(1, 1, 0, 0, 1) =$$

$$\mathbb{P}(1, 0, 1, 0, 1) =$$

$$\mathbb{P}(1, 0, 1, 1, 0) =$$

Exchangeability

Example (continued)

Starting with conditional independence (conditioning on θ) and mixing over uncertainty in θ gives

$$\begin{aligned}\mathbb{P}(X_1 = x_1, \ldots, X_n = x_n) &= \int \left[\prod_{i=1}^{n} p(x_i \mid \theta)\right] \Pi(\mathrm{d}\theta) \\ &= \int \left[\prod_{i=1}^{n} p(x_{\sigma(i)} \mid \theta)\right] \Pi(\mathrm{d}\theta) \\ &= \mathbb{P}\left(X_1 = x_{\sigma(1)}, \ldots, X_n = x_{\sigma(n)}\right),\end{aligned}$$

an exchangeable marginal distribution.

What about the reverse?

That is... Can we start by assuming only exchangeability and get the integral representation above?

de Finetti's theorem

0-1 Representation Theorem (de Finetti, 1935)

The infinite sequence of binary random variables $X_1, X_2, \ldots$ is exchangeable iff there exists a distribution function Π such that for all n, the joint probability distribution of $(X_1, \ldots, X_n)$ has a mixture model representation

$$\mathbb{P}(X_1 = x_1, \ldots, X_n = x_n) = \int \left[\prod_{i=1}^{n} \theta^{x_i} (1-\theta)^{1-x_i} \right] \Pi(\mathrm{d}\theta)\,,$$

where $\Pi(\cdot)$ is the distribution function such that

$$\Pi(\theta) = \lim_{n\to\infty} \mathbb{P}\left(\frac{1}{n} \sum_{i=1}^{n} X_i \leq \theta \right),$$

which is referred to as de Finetti's measure.

See Heath & Sudderth (1976) for a proof.

de Finetti's theorem

The representation theorem extends beyond binary random variables (see Schervish, 1995; Diaconis, 1988)

Implications:

- Assume exchangeability $\rightarrow$ mixture representation of conditional iid distributions with "parameters" and a "prior"
- Corollary:

$$\begin{aligned}\mathbb{P}(X_{n+1} = x_{n+1}, \ldots, X_{n+m} = x_{n+m} \mid x_1, \ldots, x_n) \\ = \int \left[\prod_{i=n+1}^{n+m} p(x_i \mid \theta) \right] \Pi(\mathrm{d}\theta \mid x_1, \ldots, x_n)\end{aligned}$$

- Hierarchical models are typically built on conditional independence

Hierarchical dependence

iid $\Rightarrow$ exchangeable

exchangeable $\not\Rightarrow$ iid

Hierarchical dependence

Note that exchangeability is a *weaker* assumption than ~~independence~~ iid.

In fact, exchangeability does *not* imply independence.

The mixture structure *induces* dependence among $\{X_i\}$.

Bayesians often refer to this dependence from hierarchical modeling as **borrowing strength**.

- Priors and hierarchical structure provide built-in **regularization** and **shrinkage**

Hierarchical dependence

Example (continued)

Free-throw model:

$$
\begin{aligned}
y_i \mid \theta_i &\overset{\text{ind}}{\sim} \text{Binom}(n_i, \theta_i)\,, \quad i = 1, \ldots, n, \\
\theta_i \mid \alpha, \beta &\overset{\text{iid}}{\sim} \text{Beta}(\alpha, \beta)\,, \quad i = 1, \ldots, n, \\
(\alpha, \beta) &\sim \Pi_{\alpha,\beta}\,.
\end{aligned}
$$

Where are we assuming exchangeability?

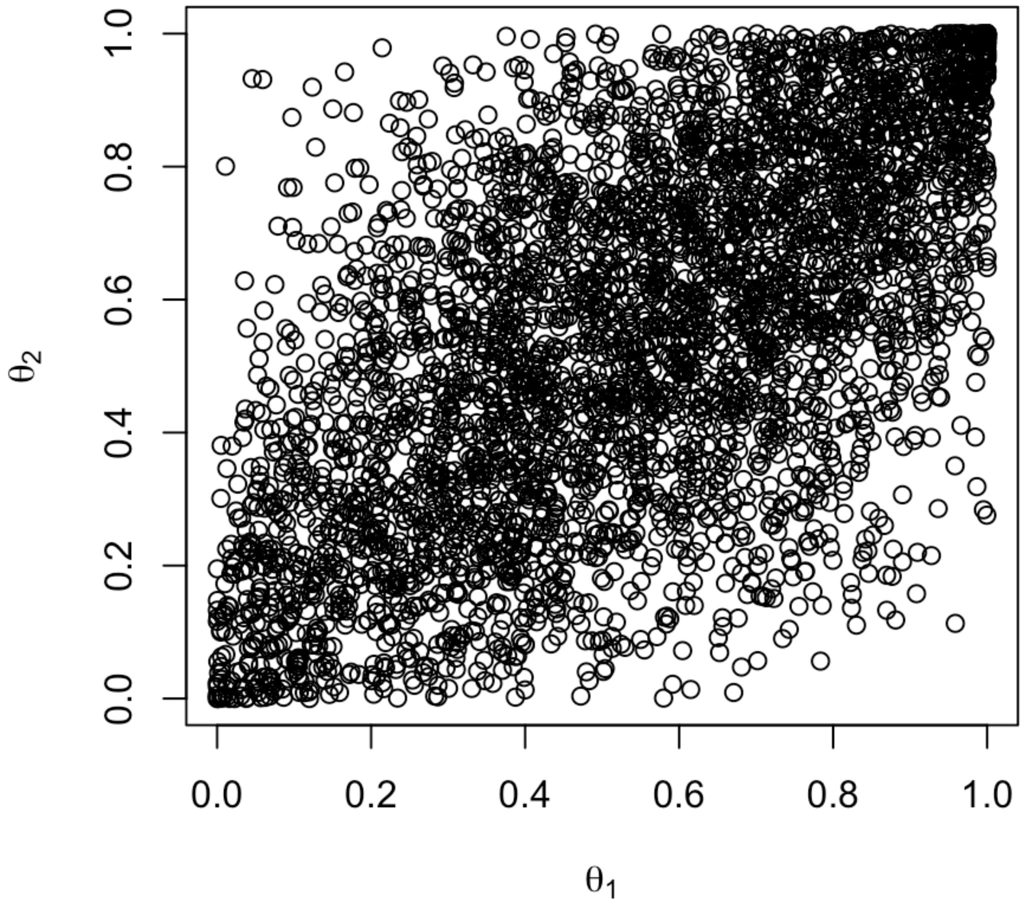
at the θ_i level

Hierarchical dependence

Example (continued)

Free throws: Verify dependence among the $\{\theta_i\}$ in the marginal prior.

▶ Code



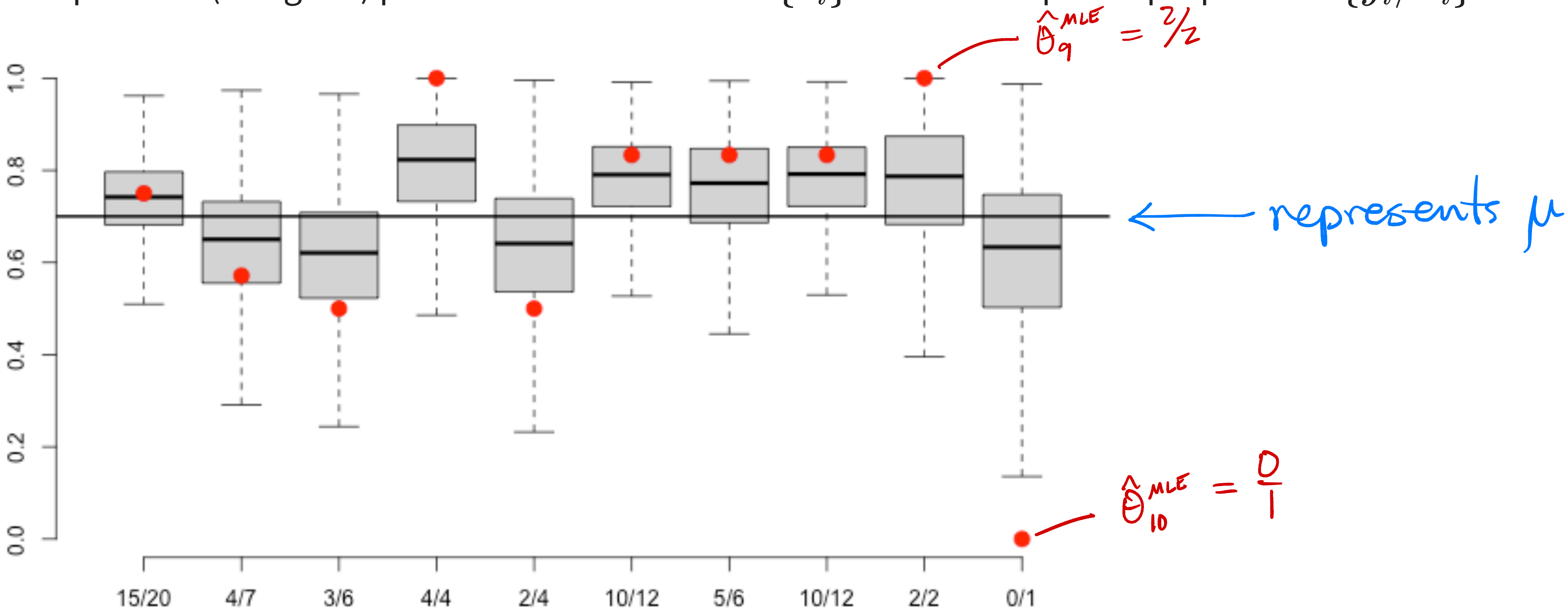
1.0
0.8
0.6
0.4
0.2
0.0
θ_2
0.0
0.2
0.4
0.6
0.8
1.0
θ_1

Hierarchical dependence

Example (continued)

Free throws: (# made / # attempts)

Compare the (marginal) posterior distributions of $\{\theta_i\}$ with the empirical proportions $\{y_i/n_i\}$.



Hierarchical dependence

Example (continued)

Free throws: (# made / # attempts)

Which of Player 4 (4/4) or Player 6 (10/12) do you think is a better free-throw shooter?

Exchangeability

What might “break” an assumption of exchangeability?

– Knowing something about the individual players like covariates x_i

What could we do about it?

– Could jointly model the response and covariates together as conditionally iid

$$(y_i, x_i) \mid \theta \overset{iid}{\sim} F$$

Hierarchical Normal model

Normal model

The hierarchical normal model is useful because

- Marginal distributions / covariances are analytically available
- Amenable to theoretical study
- Forms the basis for common regression / mixed models
- Applies broadly because of Central Limit Theorem

Normal model

Hierarchical means ANOVA model

Consider the model

$$
\begin{aligned}
y_{i,j} \mid \theta_j,\, \sigma^2 &\overset{\text{ind}}{\sim} \text{Normal}(\theta_j,\, \sigma^2)\,, \quad i = 1, \ldots, n_j, \quad j = 1, \ldots, J, \\
\theta_j \mid \mu,\, \tau^2 &\overset{\text{iid}}{\sim} \text{Normal}(\mu,\, \tau^2)\,, \quad j = 1, \ldots, J, \\
(\mu, \tau^2, \sigma^2) &\sim \pi(\mu, \tau^2, \sigma^2)\,.
\end{aligned}
$$

Potential uses?

Normal model

Hierarchical means ANOVA model

8 schools

$$
\begin{aligned}
y_{i,j} \mid \theta_j, \sigma^2 &\overset{\text{ind}}{\sim} \text{Normal}(\theta_j, \sigma^2)\,, \quad i = 1, \ldots, n_j, \quad j = 1, \ldots, J, \\
\theta_j \mid \mu, \tau^2 &\overset{\text{iid}}{\sim} \text{Normal}(\mu, \tau^2)\,, \quad j = 1, \ldots, J, \\
(\mu, \tau^2, \sigma^2) &\sim \pi(\mu, \tau^2, \sigma^2)\,.
\end{aligned}
$$

Interpretation of θ_j: Mean score in school j

Interpretation of σ: Student to student variability within school

Interpretation of μ: Overall mean score

Interpretation of τ: School to school variability (of means)

Normal model

Hierarchical means ANOVA model

$$\begin{aligned} y_{i,j} \mid \theta_j, \sigma^2 &\overset{\text{ind}}{\sim} \text{Normal}(\theta_j, \sigma^2)\,, \quad i = 1, \ldots, n_j, \quad j = 1, \ldots, J, \\ \theta_j \mid \mu, \tau^2 &\overset{\text{iid}}{\sim} \text{Normal}(\mu, \tau^2)\,, \quad j = 1, \ldots, J. \end{aligned}$$

What happens as $\tau \to \infty$?

Eight separate indep analyses

Zero borrowing of information

What happens as $\tau \to 0$?

all $\theta_j \to \mu$: one analysis

"completely pooled"

Normal model

Hierarchical means ANOVA model

$$y_{i,j} \mid \theta_j,\, \sigma^2 \overset{\text{ind}}{\sim} \text{Normal}(\theta_j,\, \sigma^2)\,, \quad i = 1, \ldots, n_j, \quad j = 1, \ldots, J,$$
$$\theta_j \mid \mu,\, \tau^2 \overset{\text{iid}}{\sim} \text{Normal}(\mu,\, \tau^2)\,, \quad j = 1, \ldots, J,$$
$$(\mu, \tau^2, \sigma^2) \sim \pi(\mu, \tau^2, \sigma^2)\,.$$

Find:

1. The marginal sampling model of $y_{i,j} \mid (\sigma^2, \mu, \tau^2)$.
2. $\text{Cov}(y_{i,j}, y_{i',j} \mid \theta_j, \sigma^2, \mu, \tau^2)$. $= 0$
3. $\text{Cov}(y_{i,j}, y_{i',j} \mid \sigma^2, \mu, \tau^2)$. $= \text{Cov}(\mu + \nu_j + \varepsilon_{ij}, \mu + \nu_j + \varepsilon_{i'j}) = 0+0+0+0+\tau^2+0+0+0+0 = \tau^2$
4. $\text{Cov}(y_{i,j}, y_{i',j'} \mid \theta_j, \theta_{j'}, \sigma^2, \mu, \tau^2)$. $= 0$
5. $\text{Cov}(y_{i,j}, y_{i',j'} \mid \sigma^2, \mu, \tau^2)$.

$y_{ij} = \theta_j + \varepsilon_{ij}$
$= \mu + \nu_j + \varepsilon_{ij}$

$y_{ij} \mid \sigma^2, \mu, \tau^2 \sim N(\mu, \tau^2 + \sigma^2)$

$y_{ij} = \underbrace{\theta_j} + \varepsilon_{ij} \sim N(0, \sigma^2)$
$= \mu + \nu_j \sim N(0, \tau^2)$

$y_{i'j'} = \theta_{j'} + \varepsilon_{i'j'}$
$= \mu + \nu_{j'}$

$\text{Cov}(y_{ij}, y_{i'j'}) = \text{Cov}(\mu + \nu_j + \varepsilon_{ij},\ \mu + \nu_{j'} + \varepsilon_{i'j'}) = 0$

Normal model

Hierarchical means ANOVA model

$$
\begin{aligned}
y_{i,j} \mid \theta_j,\, \sigma^2 &\overset{\text{ind}}{\sim} \text{Normal}(\theta_j,\, \sigma^2)\,, \quad i = 1, \ldots, n_j, \quad j = 1, \ldots, J, \\
\theta_j \mid \mu,\, \tau^2 &\overset{\text{iid}}{\sim} \text{Normal}(\mu,\, \tau^2)\,, \quad j = 1, \ldots, J, \\
(\mu, \tau^2, \sigma^2) &\sim \pi(\mu, \tau^2, \sigma^2)\,.
\end{aligned}
$$

Assume $\mu \sim \text{Normal}(m_0, V_0), \tau^2 \sim \text{Inv-Gamma}(a_\tau, b_\tau), \sigma^2 \sim \text{Inv-Gamma}(a_\sigma, b_\sigma)$.

1. Write the full joint distribution of everything.
2. Derive the complete conditionals for a Gibbs sampler.
3. What other priors for τ^2 and σ^2 might we choose?

$\theta_j \mid \cdots \sim N(m_1, v_1)$ from cond. conjugate normal known variance

using only y_{ij} from school j and $m_0 = \mu$, $V_0 = \tau^2$.

Normal model

Hierarchical means (and variances) ANOVA model

A natural extension:

$$
\begin{aligned}
y_{i,j} \mid \theta_j,\, \sigma_j^2 &\overset{\text{ind}}{\sim} \text{Normal}(\theta_j,\, \sigma_j^2)\,, \quad i = 1, \ldots, n_j, \quad j = 1, \ldots, J, \\
(\theta_j\,, \sigma_j^2) \mid \mu,\, \tau^2,\, \psi &\overset{\text{iid}}{\sim} \text{Normal}(\mu,\, \tau^2) \times \pi_{\sigma^2}(\psi)\,, \quad j = 1, \ldots, J, \\
(\mu, \tau^2, \psi) &\sim \pi(\mu, \tau^2, \psi)\,.
\end{aligned}
$$

Choices for $\pi_{\sigma^2}(\psi)$?

$$
\left.
\begin{aligned}
&\mu \sim \text{Normal} \\
&\tau^2 \sim \text{Inv-Gamma} \\
&\sigma^2 \sim \text{Inv-Gamma}
\end{aligned}
\right\}
$$
These choices would result in ALL full conditionals being conjugate

Hierarchical model in PPL

Because $y_j = (y_{1j}, y_{2j}, \dots, y_{n_j j})$ may be of different lengths across $j = 1, \dots, J$, it is usually a good idea to pass one long y vector to the PPL together with a vector indicating group (for example, school) membership.

For example $grp = (1, 1, \dots, 1, 2, 2, \dots, 2, 3, \dots, 3, 4, \dots, 4, 5, \dots, 5, 6, \dots J)$.

The length of grp is $n_1 + n_2 + \dots + n_J = N$.

Hierarchical Model in PPL

Then, in the model spec code:

```
for i in 1:N{
    y[i] ~ Normal(θ[grp[i]], σ)
}
```

OR even better,

```
for  i in 1:N {
    theta_vec[i] = theta[grp[i]]
}
y ~ Normal(theta_vec, σ)
```

* because distributional assignments are less efficient in loops than deterministic assignments in loops.

* Vectorized distrib. assignment

Model structure as latent variables

Latent model selection

Hierarchical models can also be used to express discrete model selection

$$
\begin{aligned}
\boldsymbol{y} \mid \boldsymbol{\theta}^{(m)}, Z = m &\sim f_m(\boldsymbol{y} | \boldsymbol{\theta}^{(m)}) \,, \\
\boldsymbol{\theta}^{(m)} \mid Z = m &\sim \pi_m(\boldsymbol{\theta}^{(m)}) \,, \\
Z &\sim \pi_0(z) \,,
\end{aligned}
$$

where Z is a latent variable indicating the active model among M choices.

Warning

We can easily evaluate the posterior for Z if π_m and f_m are conjugate.

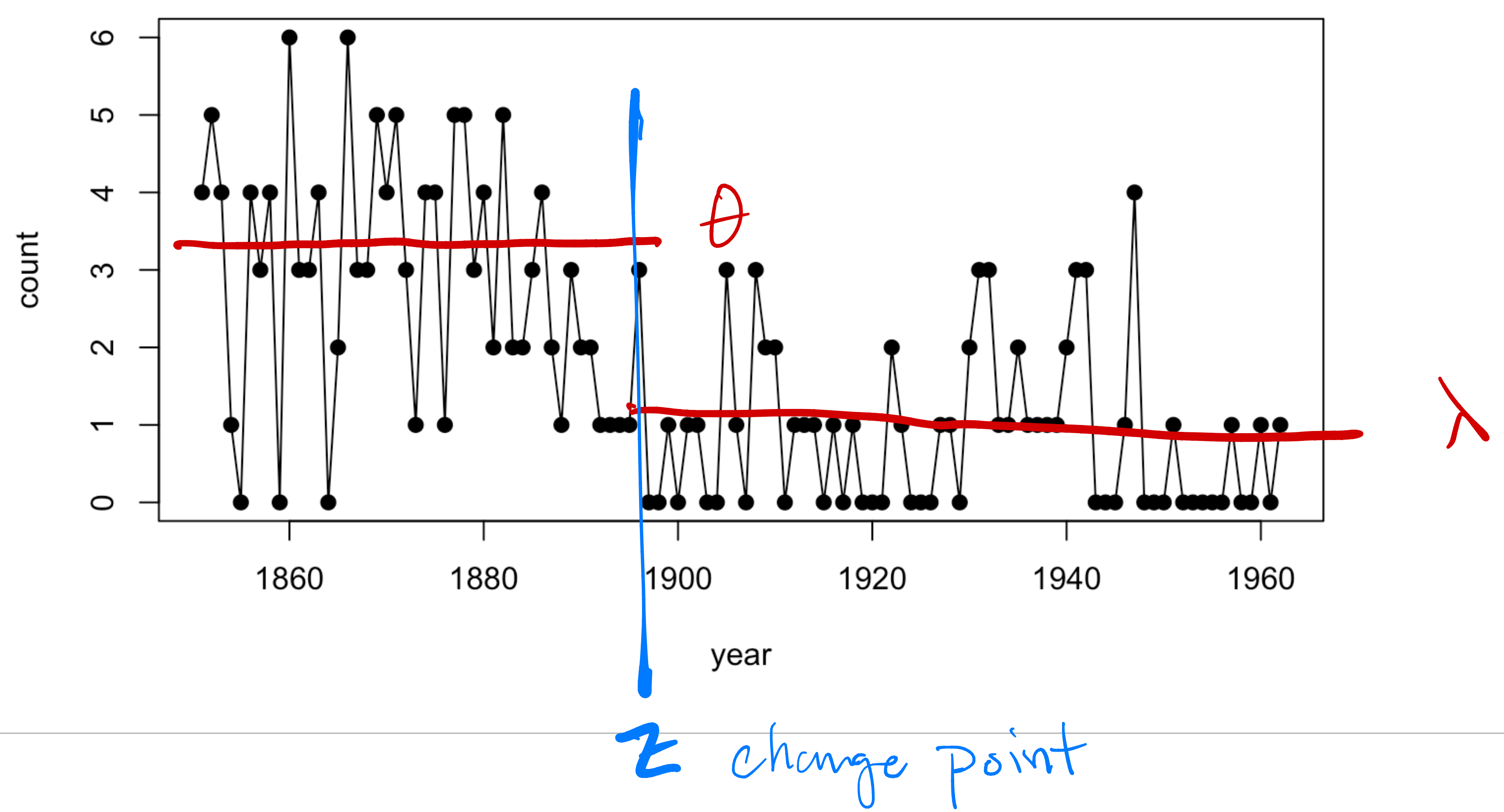
Otherwise, specialized MCMC (such as Reversible Jump) may be necessary to sample $\boldsymbol{\theta}^{(m)}$, which may change dimension with m.

Latent model selection

Example

Carlin et al., (1992) analyze a time series of coal-mining disasters in Great Britain between 1851 and 1962 (reported by Maguire et al., 1952 and Jarrett, 1979; available in boot CRAN package).

▶ Code


count
year
θ
λ
z change point

Latent model selection

Example (continued)

Carlin et al., (1992) propose to model the counts as Poisson with a change point:

$$
\begin{aligned}
y_t \mid \theta, Z &\stackrel{\text{iid}}{\sim} \text{Poisson}(\theta)\,, \quad t = 1, \ldots, Z, \\
y_t \mid \lambda, Z &\stackrel{\text{iid}}{\sim} \text{Poisson}(\lambda)\,, \quad t = Z+1, \ldots, T, \\
\theta \mid a_\theta, \gamma_\theta &\sim \text{Gamma}(a_\theta, \overset{\text{rate}}{\cancel{\text{scale}}} = \gamma_\theta)\,, \\
\lambda \mid a_\lambda, \gamma_\lambda &\sim \text{Gamma}(a_\lambda, \overset{\text{rate}}{\cancel{\text{scale}}} = \gamma_\lambda)\,, \\
Z &\sim \text{Uniform}(\{0, 1, 2, \ldots, T\})\,,
\end{aligned}
$$

where $T = 112$. They complete the model with gamma priors on γ_θ and γ_λ.

Can you spot any potential problems with model identifiability? How might we address them?

Latent model selection

Example (continued)

Carlin et al., (1992) propose to model the counts as Poisson with a change point:

$$
\begin{aligned}
y_t \mid \theta, Z &\overset{\text{iid}}{\sim} \text{Poisson}(\theta)\,, \quad t = 1, \ldots, Z,\\
y_t \mid \lambda, Z &\overset{\text{iid}}{\sim} \text{Poisson}(\lambda)\,, \quad t = Z+1, \ldots, T,\\
\theta \mid a_\theta, \gamma_\theta &\sim \text{Gamma}(a_\theta, \overset{\text{rate}}{\cancel{\text{scale}}} = \gamma_\theta)\,,\\
\lambda \mid a_\lambda, \gamma_\lambda &\sim \text{Gamma}(a_\lambda, \overset{\text{rate}}{\cancel{\text{scale}}} = \gamma_\lambda)\,,\\
Z &\sim \text{Uniform}(\{0, 1, 2, \ldots, T\})\,.
\end{aligned}
$$

1. Write the full joint distribution of everything.
2. Derive the complete conditionals for a Gibbs sampler.
3. Derive a collapsed Gibbs update for Z (integrating out θ and λ).

Likelihood : $f(y \mid \theta, \lambda, Z) = \prod_{t=1}^{Z} \left[\text{Pois}(y_i \mid \theta)\right] \prod_{t=Z+1}^{T} \text{Pois}(y_t \mid \lambda)$

FC for Z: $\propto f(y \mid \theta, \lambda, Z) \frac{1}{T+1}$

Latent model selection

Example (continued)

Carlin et al., (1992) propose to model the counts as Poisson with a change point:

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\begin{aligned}
y_t \mid \theta, Z &\overset{\text{iid}}{\sim} \text{Poisson}(\theta)\,, \quad t = 1, \dots, Z, \\
y_t \mid \lambda, Z &\overset{\text{iid}}{\sim} \text{Poisson}(\lambda)\,, \quad t = Z+1, \dots, T, \\
\theta \mid a_\theta, \gamma_\theta &\sim \text{Gamma}(a_\theta, \overset{\text{rate}}{\cancel{\text{scale}}} = \gamma_\theta)\,, \\
\lambda \mid a_\lambda, \gamma_\lambda &\sim \text{Gamma}(a_\lambda, \overset{\text{rate}}{\cancel{\text{scale}}} = \gamma_\lambda)\,, \\
Z &\sim \text{Uniform}(\{0, 1, 2, \dots, T\})\,.
\end{aligned}
$$

The model above is *far* from the only choice available to us.

What other model structures might make sense?